

SAFETYAI V2.0 PITCH & DEMO PLAYBOOK

Industrial Incident Orchestration Simulator & Command Center

Category	Predictive Safety / Worker Geospatial Analytics
Hackathon Theme	ET AI Hackathon 2026 - Industrial Intelligence
Presentation Time limit	3 to 4 Minutes Total
Backend Architecture	FastAPI Event Engine + Time-Series Historical Streams
Frontend Framework	React + Recharts Responsive Viewport

Playbook Purpose: This guide bridges the engineering implementation and the final pitch flow. It provides a slide-by-slide script, synchronized physical screen actions (hovers/clicks), time benchmarks, and visual anchors to ensure the judges grasp the compound hazard reasoning values.

■ DEMO WALKTHROUGH & PRESENTATION SCRIPT

Beat 1: The Hook & SCADA's Silent Disasters (Time: 0:00 - 0:30)

Goal: Establish the human cost and the SCADA technical boundary gap.

■ ■ PITCH SPEECH:

"Judges, thank you. Indian heavy industries continue to pay a devastating human cost. During the Visakhapatnam Coke Oven gas leak, telemetry data existed: elevated pressure was recorded. But traditional SCADA systems remained silent because levels were below the static threshold. SCADA is blind to work permits and rosters. SafetyAI connects these siloed streams, acting as an intelligence layer to predict hazards 33 minutes early. Let's launch the console."

■ MOUSE & KEYBOARD ACTIONS:

1. Show the Onboarding Split comparison splash screen.
2. Hover the mouse cursor over the 'Traditional SCADA' warning box, then the 'SafetyAI' predictive box.
3. Click the 'Launch Intelligence Console' button to transition.

Beat 2: Onboarding & Scenario Selection (Time: 0:30 - 0:45)

■ ■ PITCH SPEECH:

"We are entering the main Control Center Selector. SafetyAI hosts a parameter-driven configuration registry supporting five scenarios, from Ammonia leaks to Boiler overpressures. We can select presets like 'Operational Drift' or 'Critical Escalation' to simulate varying severity patterns. Let's run a Coke Oven Gas Leak under Operational Drift."

■ MOUSE & KEYBOARD ACTIONS:

1. On the scenario registry page, highlight the Preset Toggles (Operational Drift vs. Critical Escalation).
2. Hover over the ****Ammonia Storage Leak**** and ****Oxygen Deficiency**** cards to show support for multiple gases.
3. Click the ****Coke Oven Gas Leak**** card to load the active cockpit.

Beat 3: Cinematic Dashboard Overview (Time: 0:45 - 1:15)

■■ PITCH SPEECH:

"Notice the unified CommandCenter dashboard. To maximize user context, it sits on a single, strict viewport. On the left: a real-time geospatial layout of the plant's eight monitored zones, overlaid with current permits. Below it: our State-Aware Context HUD mapping shift fatigue profiles. On the right: the Predictive Trend Chart and RAG memory logs. Standard telemetry shows safe, green limits. But as operations begin to drift..."

■ MOUSE & KEYBOARD ACTIONS:

1. Hover over the green circles on the Plant Map.
2. Hover over the ****Context HUD**** showing 'Evening Shift' and 'Workers: 8'.
3. Point cursor to the green baseline on the ****Predictive Trend Chart****.

Beat 4: Anomaly Drift & The Predictive Curve (Time: 1:15 - 1:50)

■■ PITCH SPEECH:

"Watch the telemetry. The system transitions from SAFE to WATCH. While traditional SCADA alarms remain silent since the level is far below the 50 ppm limit, SafetyAI's trend algorithms extrapolate the curve, predicting a threshold breach in exactly 33 minutes. The countdown starts."

■ MOUSE & KEYBOARD ACTIONS:

1. Observe telemetry rising on the chart. Point cursor to the dashed blue ****AI Forecast Line**** and the shaded ****Confidence Interval Band****.
2. Point to the ****Breach Predicted**** red warning badge appearing above the chart.
3. Highlight the ****33 Minutes Remaining**** countdown in the main header.

Beat 5: Context Correlation & Recommendation Plans (Time: 1:50 - 2:20)

■ ■ PITCH SPEECH:

"At 10 minutes, the risk escalates. SafetyAI fuses context: it flags a welder's active Hot Work permit inside the leak zone during a high-fatigue shift handover. The RAG memory agent correlates historical records, finding a 91% match with a previous valve incident. The system immediately compiles candidate mitigation strategies, calculating cost-benefit trade-offs for Plans A, B, and C."

■ MOUSE & KEYBOARD ACTIONS:

1. Point to the ****Context HUD**** where the permit card highlights 'SPARK IGNITION RISK'.
2. Point to the ****Historical Match Card**** showing 'Incident #23 similarity: 91%'.
3. Scroll slightly down to the ****Reasoning Narrator Feed**** to highlight logs: 'PlannerAgent: Plan A, B, C generated.'

Beat 6: Consensus Debate & Approval Gate (Time: 2:20 - 3:00)

■ ■ PITCH SPEECH:

"When risk crosses 75%, SafetyAI launches the Consensus Debate. Risk Agent, Cost Agent, and Safety Agent weigh trade-offs. The Judge Agent recommends Plan A: Selective Welding Permit Suspension. The dashboard pauses, locking active operations in the Human Approval Gate. Interventions require a safety officer's sign-off. Let's approve the action."

■ MOUSE & KEYBOARD ACTIONS:

1. Highlight the three side-by-side agent arguments in the ****Debate Panel****.
2. Highlight the blinking red banner at the top: 'AUTONOMOUS ACTIONS PAUSED - AWAITING APPROVAL'.
3. Click ****Approve Plan A**** inside the banner.



Beat 7: Recovery Loop & Outcome Scorecard (Time: 3:00 - 3:30)

■ PITCH SPEECH:

"Upon approval, the welding permit is instantly suspended. Air ventilation capacity ramps to 100%. Watch the chart: telemetry falls, and we transition into the CLOSED phase. SafetyAI overlays a dynamic compliance summary: 8 workers protected, **20.4L+** downtime avoided, and generates our structured incident audit KPI metrics. SafetyAI connected the reasoning, preventing the disaster. Thank you."

■ MOUSE & KEYBOARD ACTIONS:

1. Observe the ****Permit PTW log**** flashing red overlay: 'SUSPENDED BY SAFETY OFFICER'.
2. Watch the chart values decline and the overall index return to green.
3. Highlight the overlay table showing workers saved and hours avoided.
4. Click the ****Download Compliance Audit Summary**** button to complete the flow.

■ ARCHITECTURE REVIEW NOTES

During Q&A;, utilize these structured responses to address technical questions from the judges:

Question	Target Response Beat
How does the prediction model work?	It utilizes trend extrapolation. In production, this scales to LSTM Autoencoders factoring in wind direction and air pressure.
How is compliance validated?	Statically verified checks against Hot Work overlap. Production scales to dynamic Cypher queries on Neo4j Graph Databases mapping Indian Factory Acts.
How are metrics calculated?	Scorecards compute downtime cost from hourly rates multiplied by safety lead hours gained. Workers protected evaluates active permit crew counts.